

Dear Students,

Here is Assignment-4, submission deadline: December 02, 2025.

Allowed programming language: C/C++.

You should submit -

1. the source code (C/C++) file
2. Input File (contains the string on which the pattern needs to be matched)
3. Pattern matching file - contains 5 patterns in 5 lines. At least one pattern should contain ?, at least one should contain *, and at least one should contain both * and ?.
4. A shell script- running the shell script will execute your program with all the necessary input and pattern file.

Combine all these files in a zip file and name it as per your roll number. Google form link for the submission is - <https://forms.gle/Z9NT56wZwDeEenb79>

Build a Generalized Pattern-Matching Engine (GPME)

Create a module with the following features:

Supports pattern searching for:

Exact match

Case-insensitive match

Multi-pattern match using KMP repeatedly (patterns ≤ 20)

Supports optional wildcards:

? → matches exactly one character

* → matches zero or more characters

Implement wildcard support without using regex; create your own Dynamic Programming or finite automaton-based logic.

For each search mode, output:

Number of matches

Match positions

Total comparisons